EXECUTIVE SUMMARY

Use the risk model to price loans and set aside reserves, not to decline more borrowers

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how often the riskiest tenth of borrowers defaulted, compared with the safest tenth

$260m vs $274m

predicted vs actual credit losses on 2015 loans, after the yearly recalibration banks apply

About 5%

the point beyond which declining more borrowers lost money

The question

Which loan applicants will repay, and can a model built only from application data outperform the lender's own risk grades?

The evidence

546,000 finished 36-month loans. The model was tested on 282,787 loans from 2015, worth $3.62bn, that it never saw during development.

The answer

- The model ranks borrowers better than the lender's own grade.
- The lender's interest rates already cover most of the risk, so declining many more borrowers would reduce returns.
- The model's greatest value is in pricing and in forecasting losses.

THE SITUATION

Lending mistakes are expensive, and simple rules leave the hardest decisions open

About $1,800

of expected profit lost, on average, when a good borrower is declined

About $5,000

lost, on average, when an approved borrower does not repay

64%

of 1.61 million applications declined by four simple policy rules, such as a minimum credit score

Why it matters

Lenders receive far more applications than people can review. Rules handle the clear-cut cases cheaply, but the remaining 36% need a sharper judgement of who will repay.

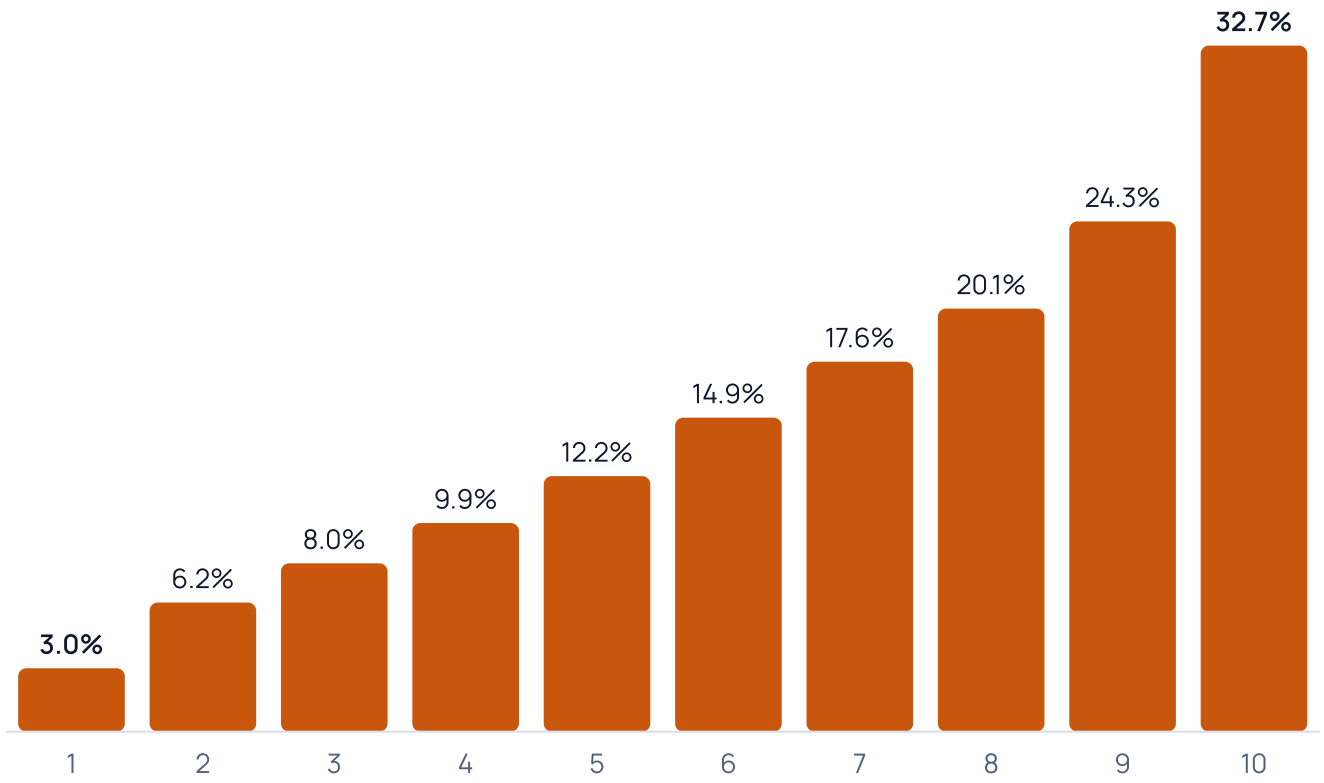
What was done

1. Cleaned and combined 29.9 million raw rows into 1.61 million comparable applications.
2. Built a model that estimates each borrower's chance of not repaying, leaving out the lender's own grade so the two could be compared fairly.

FINDING 1

The model separates risky from safe borrowers, and spots risk the lender's grade misses

Share of 2015 borrowers who did not repay



Borrowers grouped into tenths by the model, from 1 (safest) to 10 (riskiest)

- **Better ranking.** On a standard ranking score (Gini, where higher is better), the model scored 0.385 against 0.357 for the lender's grade.
- **New information.** Adding the model to the grade raised the score from 0.360 to 0.407.
- **Explainable.** Credit score matters most, followed by recently opened accounts, loan size relative to income, and debt relative to income.

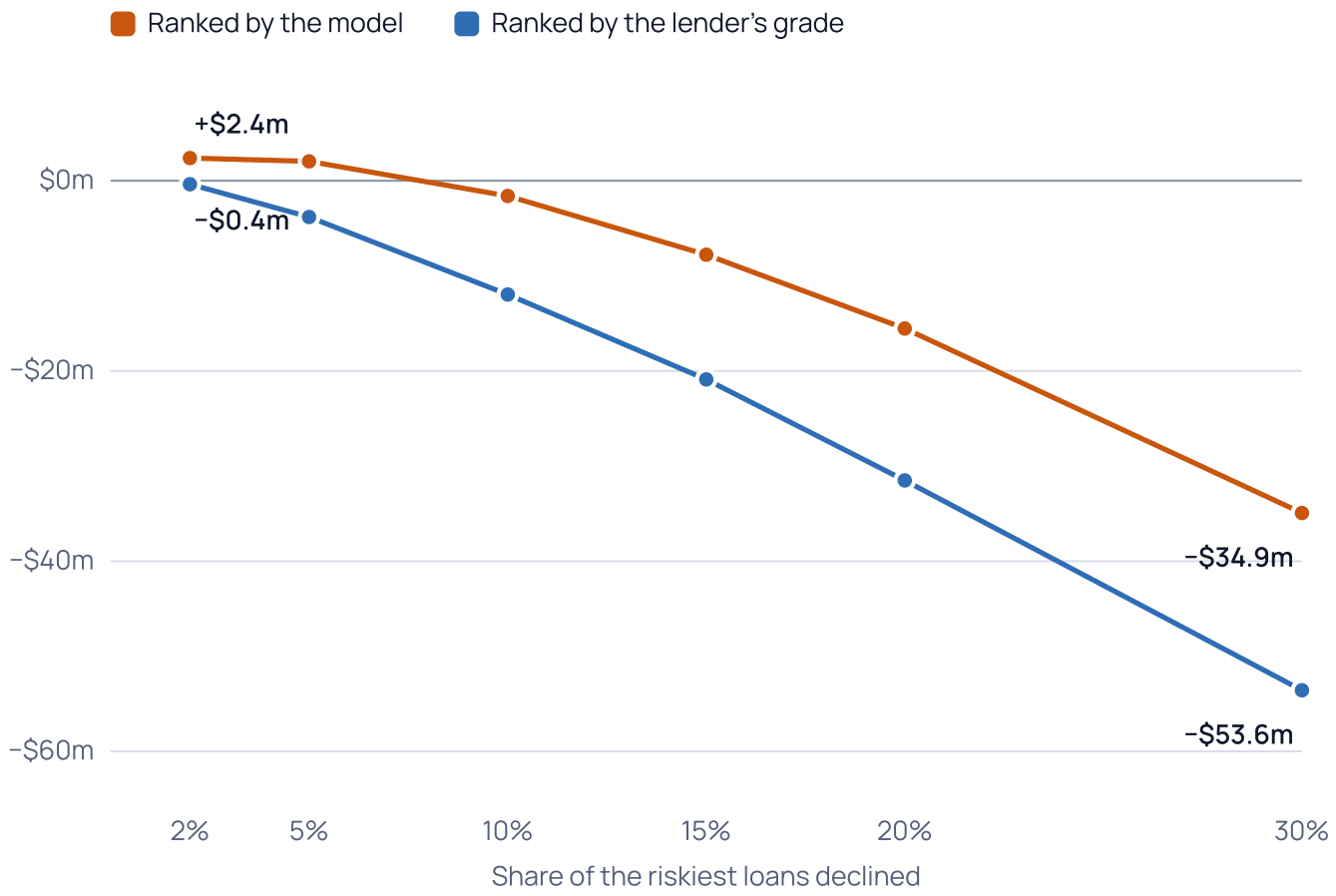
SO WHAT

The riskiest tenth defaulted at 32.7% and the safest at 3.0%. Differences this large can support sharper pricing.

FINDING 2

Declining more than about 5% of borrowers loses money, because interest rates already cover most of the risk

Change in net result on 2015 loans, by share of riskiest loans declined



- **A small gain, then losses.** Declining the riskiest 2% by the model's ranking added $2.4m. At 10%, the change was already negative.
- **Better at every level.** The model's ranking beat the grade each time, for example −$15.5m against −$31.5m at 20%.

SO WHAT

Declined loans include many good borrowers whose interest pays for the defaults. A stricter cut-off is not the main source of value.

RECOMMENDATION

Three moves for a lender, with known limits to address before any rollout

What to do

1. **Keep the rules for clear-cut cases**
   Use the model for the applications they leave open.
2. **Price and reserve using the model**
   Use it to set rates and forecast losses, and recalibrate it every year.
3. **Decline only the riskiest few percent**
   Stay at or below about 5%, and give the top two reasons for every decline, as US lending rules require.

Next step

Re-test on the lender's own loans, and estimate how rejected applicants would have performed.

Limits to keep in mind

- The model only learned from approved borrowers, as rejected applicants have no repayment record.
- It covers 36-month loans only. 60-month loans need their own model.
- Default rates rose from 12.3% in 2013 to 14.9% in 2015, so loss forecasts drift without recalibration.
- Loss forecasts use one average recovery rate, not an estimate for each loan.
- Net results exclude servicing fees and funding costs.
- Fairness checks were limited to excluding location data, as the data holds no protected attributes.